

Harrison Jansma

MACHINE LEARNING ENGINEER · GENERATIVE AI & REAL-TIME NLP SYSTEMS

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Applied ML engineer who takes ambiguous problems to production and then owns them. Six years at Capital One across credit risk, real-time NLP, and generative AI – most recently creating the GenAI scope for a resource-constrained call-center team rather than inheriting it, then building the systems myself. Learning speed is what I lean on: I step outside my role – a partner team's stack, deployment tuning, product management – to make things viable. Shipping solo products outside work too, with coding agents on the keystrokes.

HOW I OPERATE

Scope taken, not assigned — built the call center's GenAI charter from nothing in seven months; now lead a ~10-person cross-functional team with no line authority.

Learns fast, in the deep end — picked up a partner team's unfamiliar orchestration stack in days and shipped production code in their repo.

Every hat — model, streaming pipeline, deployment, annotation program, evals, interim PM, executive storytelling. Whatever the product needed to exist.

Delivers under ambiguity — carried a flagship pilot solo after the team lost its manager and its other scientist, with no engineering resources.

EXPERIENCE

Capital One — Plano, TX

Apr 2020 – Present

Data Science Manager (2024–Present) · Principal Data Scientist (2022–2024) · Sr. Associate, Data Science (2021–2022) · Data Process Manager (2020–2021)

GENERATIVE AI FOR THE CONTACT CENTER — CURRENT FOCUS

- Own delivery of GenAI to the auto-finance call centers – a scope I created, not one I was handed. Two capabilities are live in the agent UI today: real-time call summarization and RAG retrieval over live transcripts.
- Technical and strategy lead for a ~10-person cross-functional team (2 PMs, a software engineer, a data engineer, 5 annotators) organized without line authority – plus two applied-research pods and a 3-engineer pipelining team I negotiated for when the roadmap outran my hands.
- Set direction with VP- and EVP-level stakeholders and sequenced builds through mandatory model-risk governance, so regulation slowed the paperwork, not the shipping.
- Delivered real-time call summarization end to end and hands-on: a Kubernetes-deployed FastAPI service over an internally hosted LLM, plus the Prefect/Kafka streaming pipeline – written inside a partner team's codebase after learning their stack in days. Phased rollout live; full rollout late 2026, with meaningful projected savings in after-call work.
- Built the evaluation harness the pilot had to clear: human-annotated calls plus LLM auto-judging for factuality and completeness.
- Designed and shipped to pilot the RAG document-suggestion system: live transcript segments matched against a vector database of policy documents in under a second, surfacing clickable suggestions as the agent talks. Built the weekly pipeline behind it – an internal policy site turned into a clean embeddable corpus, the first data linkage between calls and the policies governing them – and swept dozens of ranking algorithms in a day with agentic coding when agents flagged jumpy suggestions.
- Built and validated a GPT-based complaint-detection model that substantially outperforms the legacy classifier on real escalations – solo analysis and production implementation, recognized with an internal Diamond award. Pilot planned August 2026.

REAL-TIME NLP — FROM ZERO TO PRODUCTION

- Architected and shipped the first real-time model in any Capital One call center: live detection of frustrated customers with sub-second alerting, now running across all of Auto and Card at roughly 100,000 calls a day.
- Built the data foundation when none existed – a two-month utterance-level annotation program with weekly agreement exercises, then months of experiments to get a transformer small and fast enough to serve live (PyTorch, Hugging Face).
- Engineered the parallel streaming inference pipelines (Kafka, DynamoDB, S3, Snowflake), tuning deployment inside the platform team's codebase to hit real-time latency, with dashboards and alerting (New Relic, PagerDuty, Splunk) keeping a non-deterministic system honest.

- Carried the pilot alone when the three-person team lost its manager and its other data scientist with no tech resources: solo-built a Slack alerting path that onboarded a new call-center team in under an hour, ran pilot tracking and unit-manager relations, and prepared analytics for CEO-level presentation – acting as interim PM throughout.

CREDIT RISK & EARLIER

- Engineered the auto-loan underwriting ML suite decisioning loans at national scale; built and owned the core decisioning model, plus a simulation engine powering reinforcement-learning credit-policy optimization.
- Led the Python implementation of automated governance reporting for the Financial Services tech org, including monthly VP-level org-health views on Snowflake – while completing an M.S. in Computer Science.

Sprint – Irving, TX

2019

Data Science Intern

- Designed a model-training framework for anomaly detection on customer billing data – 5× increase in detectable failure types; the only internship extended across an 80+ intern class.

University of Pennsylvania (via Upwork) – Remote

2018 – 2019

Machine Learning Researcher — NLP

- Built an emotion-classification model and processed millions of public social-media profiles for a population-level NLP study.

INDEPENDENT BUILDS — 2025–2026

- Solo, shipped end to end, largely by orchestrating coding agents — and the TypeScript-stack evidence my Python-first day job doesn't show.
- **Live Call Copilot** – a public, in-browser rebuild of real-time agent assist: streaming ASR, self-drafting notes, RAG over pgvector, and a sub-second frustration alert with measured latency on screen. One WebSocket per session; a Node worker fans each utterance out to notes, retrieval, and sentiment in parallel, each with its own cadence and cost guards.
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 - **Shopfolio** – a creator back-office built alone: cross-platform post scheduling, a capture-to-compose CMS, and an embeddable storefront with tracked links and Stripe billing – plus a four-layer migration-integrity system that keeps a one-person product from breaking itself.
 - **Harrison MCP** – a recruiter-facing MCP server with grounded Q&A over my track record and a fit scanner that names honest gaps as well as matches. Planned, built, and project-managed agentially, with its own eval suite.

SKILLS

Generative AI	LLM orchestration, agentic / multi-step reasoning workflows, prompt engineering, RAG, fine-tuning, transformers (BERT / DistilBERT), embeddings, GPT-based pipelines
Real-time & MLOps	Streaming inference (Kafka, AWS Lambda, DynamoDB; Flink pilot work), Prefect orchestration, FastAPI services on Kubernetes, vector databases, low-latency serving, AWS S3, Snowflake
Evaluation	LLM-as-a-judge / agent-as-a-judge, offline & online evaluation, quality-metric design, annotation programs, model observability, New Relic, PagerDuty, Splunk
Modeling	PyTorch, TensorFlow, Hugging Face, scikit-learn; NLU, sentiment & complaint detection, anomaly detection, reinforcement learning, simulation, topic modeling, clustering
Product stack	TypeScript, Node.js, Next.js (App Router), React, WebSockets, Supabase (Postgres, RLS, Auth), pgvector, Stripe, Tailwind, Playwright, pnpm monorepos (independent builds)
Languages & data	Python, SQL, C++, PySpark, Hive, Tableau
AI-first development	Agentic coding with Claude Code: semi-autonomous multi-step loops with test and review gates, context and tool design, checkpoints on irreversible steps
Leadership	Organizing cross-functional teams without line authority, partner-pod negotiation, roadmapping, VP/EVP stakeholder management, interim product management

EDUCATION

- M.S. Computer Science (Data Science) — The University of Texas at Dallas · 2018–2020
- B.B.A. Mathematics, Business Fellows — Baylor University · 2013–2017 · Magna Cum Laude, GPA 3.86, Honors Program